Innovation in Engineering

Progress Report 2

Goal: Reflect on the course regarding innovation theory and team formation.

a) Reflect on how innovation theory affected your direction in your project/teamwork?

b) Which other key components from innovation theory could in your perspective have been applied with a different outcome?

c) In the innovation theory lecture, a main point was the level of definition of the opportunity and solution space. Allocate your innovation project within this space and elaborate your choice.

Consider: Innovation approach, multidisciplinary teams, industrial partner

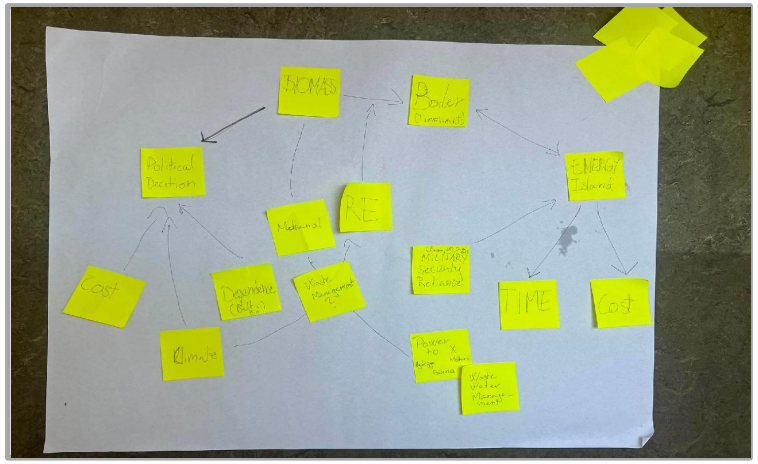
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a) Innovation theory helped us look at the challenge from different perspectives instead of focusing only on the technical solution as we always do. At the beginning, our challenge was broad and not specific. Through brainstorming and stakeholder mapping, we identified different areas such as environmental impact, economics, community development and social acceptance and other aspects. This helped us narrow down our research direction and focus on three main areas: the current energy and heating system on Bornholm, the development of the Baltic Energy Island, and the experiences and acceptance of the local community. Working in a multidisciplinary team also helped us consider the problem from different perspectives and develop a broader understanding of the challenge.

b) From my perspective, we could have done more background research at an earlier stage of the project. Since the challenge was quite broad at the beginning, having a better understanding of Bornholm's existing energy system and energy projects could help us know the key problems and develop more focused research questions earlier. We could also have involved stakeholders earlier in the process. Their perspectives could help us understand the actual needs and challenges.

c) I think our project is in a relatively open opportunity and solution space. The general opportunity is clear: Bornholm is going through a green energy transition, and there is a need to explore how the future energy and heating system can be developed in a sustainable and socially acceptable way. However, the specific problem is still being explored, and there is no concrete solution. Our research therefore focuses on understanding the current situation and identifying needs, barriers and opportunities through different stakeholders. For example, we are interviewing the municipality, BEOF, local businesses and residents. Their different perspectives may lead us to redefine the problem and explore different possible solutions.





Goal: Investigating and exploring your innovation challenge

a) How did the expanded mental model help you take different perspectives into account in your innovation challenge?

b) Which data collection methods did you use? Support why you made this specific selection and what other methods you considered, but not chosen.

c) How did the discussions on your understanding of the challenge help you to identify key questions for further investigation?

Consider: data collection methods, decision-making, and problem identification.

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a) The Expanded Mental Model helped me look at the challenge from different perspectives instead of seeing it only as a technical energy problem. We considered different aspects such as economics, environmental impact, social change, community development and ethics. We also identified different stakeholders, including local residents, BEOF and Bornholm Municipality. This helped us understand that different stakeholders may have different needs and concerns, and gave us a broader understanding of the challenge.

b) We mainly selected semi-structured interviews, street interviews and an online questionnaire. Semi-structured interviews allowed us to get detailed information from different stakeholders and also gave us the chance to ask follow-up questions based on their answers. Street interviews helped us collect opinions directly from local residents and understand their experiences and concerns. We also used an online questionnaire to supplement the street interviews and reach a larger group of local residents. By combining these methods, we could collect different types of information and understand the challenge from different perspectives. We also considered using observation as another data collection method, but we did not choose it in the end. Since our project focuses more on stakeholders' opinions, experiences and concerns, we thought interviews and questionnaires were more useful for our research. Also, because of the limited time, it would be difficult to carry out enough observations.

c) At the beginning, our challenge was very broad, so our team discussed what we already knew and what we still needed to understand. Through brainstorming, we identified knowledge gaps and different areas that needed further investigation. This helped us develop more focused research questions. For example, we decided to investigate the current energy and heating solutions on Bornholm, the development of the Baltic Energy Island, and the experiences and acceptance of the local community.





Goal: The purpose of data collection is to better understand the innovation challenge you will work with. This will help you get enough insight into the challenge to be able to identify opportunities for innovation and/or problems that stakeholders experience

a) How did the collected data provide new insights for your innovation challenge?

b) If more time was available, what additional data types would you collect and why?

c) Do you believe your current data set is sufficient to further explore your challenge? Explain why or why not.

Consider: Insight generation, resource management, and critical thinking.

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a) The collected data helped us understand the challenge from different stakeholders' perspectives. Before the interviews, we mainly focused on the old biomass boiler and how it could be replaced. However, after talking with BEOF, we found that the challenge is more complicated than we thought. For example, the limited electricity capacity on Bornholm is also a big problem, especially if more heating systems use electricity in the future. We also learned that heating prices, changing electricity prices and investment risks can affect BEOF's decisions. From the interview with Ocean Prawns, we also learned that local businesses care about affordable and accessible energy. These different perspectives helped us see the challenge in a broader way.

b) If we had more time, we would like to collect more answers from our online questionnaire (https://forms.gle/gL3Rx1t1GnEGygFt6). Since the time for data collection was quite short, we did not get as many answers as we wanted. We could keep the questionnaire open for a longer time and share it through more local channels. We would also like to interview more stakeholders, such as local residents, businesses and other organisations. From our current interviews, we found that different stakeholders care about different things. Therefore, more interviews could help us hear more opinions, compare different perspectives and understand the whole challenge better. (there is a website: www.reddit.com)

c) We think our current data is enough for us to continue exploring the challenge, but it is not enough to make a final conclusion. The interviews gave us useful information and helped us find some important problems that we did not fully understand before. However, the number of interviews and questionnaire answers is still limited. More data, especially from local residents and businesses, would help us check whether the opinions we collected are shared by more people and give us a more complete picture of the challenge.



